

Neha Verma

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Education

- **Johns Hopkins University** **Baltimore, MD**
Ph.D. in Computer Science, Advisors: Kevin Duh & Kenton Murray 08/2021 – 09/2026 (Expected)
- **Johns Hopkins University** **Baltimore, MD**
M.S.E. in Computer Science 08/2021 – 05/2023
- **Yale University** **New Haven, CT**
B.S. in Computer Science & Mathematics 08/2017 – 05/2021
– Distinction in the Major, Magna Cum Laude

Experience

- **Google DeepMind & YouTube** **Mountain View, CA**
Research Scientist Intern 05/2025 – 08/2025
Developed a novel model-merging method for continual learning [1]. U.S. Patent Pending.
- **Meta – Fundamental AI Research** **Menlo Park, CA**
Research Scientist Intern 05/2023 – 09/2023
Introduced a novel model-merging technique for Transformers that significantly reduces loss barriers between models trained from separate initializations [4].
- **Johns Hopkins University** **Baltimore, MD**
Research Assistant 08/2021 – Present
Research toward my PhD. Topics include model merging for compression [2, 3], representation learning for multilingual translation [5, 6, 9], speech translation [7, 8], and domain adaptation [10].
- **Language and Information Learning Lab (LILY) at Yale** **New Haven, CT**
Research Assistant 05/2018 – 08/2021
Undergraduate Research Assistant with Dragomir Radev. Researched structured text generation [12, 13], cross-lingual information retrieval [14], and NLP for EHRs [11].
- **CVS Health/Aetna** **Remote**
Data Science Intern 05/2020 – 08/2020
Built a model to predict adverse hyperkalemic episodes for patients with kidney disease.
- **Summer Undergrad Math Research at Yale (SUMRY)** **New Haven, CT**
Research Assistant – Advisor: Karamatou Yacoubou Djima 05/2019 – 08/2019
Researched spectral and harmonic methods for denoising placental vascular network images.

Awards and Grants

- Phi Beta Kappa 2021
- Yale College First-Year Summer Research Fellowship in the Sciences & Engineering 2018

Teaching

• Introduction to Human Language Technology <i>Teaching Assistant</i>	Johns Hopkins University <i>Fall 2025</i>
• Language and Computation <i>Course Assistant</i>	Yale University <i>Spring 2021</i>
• Fields and Galois Theory <i>Peer Tutor</i>	Yale University <i>Spring 2020, Spring 2021</i>
• Abstract Algebra <i>Course Assistant</i>	Yale University <i>Fall 2019, Fall 2020</i>
• Linear Algebra & Vector Calculus II <i>Course Assistant</i>	Yale University <i>Spring 2019</i>
• Calculus of Functions of Several Variables <i>Peer Tutor</i>	Yale University <i>Fall 2018</i>

Service and Professional Activities

Organizing

• JHU CLSP Application Support Program	<i>2022-2025</i>
• Machine Translation Marathon of the Americas (MTMA)	<i>2024</i>
• Regional NACLO @ Yale University	<i>2020-2021</i>
• Yale Association for Women in Mathematics (AWM) Chapter President	<i>2020-2021</i>

Reviewing

- **Conferences:** COLM 2026, ICML 2026; ICLR 2025, 2026; NeurIPS 2025; ACL Rolling Review 2023, 2024, 2025; AMTA 2022.
- **Journal:** Computational Linguistics (2022)

Skills

- **Programming Languages:** Python, Bash, SQL
- **Frameworks/Toolkits:** PyTorch, Jax, fairseq, Hugging Face, Dask, $\LaTeX$, Git

Publications

- [1] **N. Verma***, N. Mehta*, S. Wang, N. Zhang, A. Tsai, A. Nath, L. Wei, L. Heldt, L. Hong, E. Chi, X. Yi. "ORBIT: Preserving Foundational Language Capabilities in GenRetrieval via Origin-Regulated Merging", *preprint*, 2026.

- [2] **N. Verma**, K. Murray, K. Duh. “DOTResize: Reducing LLM Width via Discrete Optimal Transport-based Neuron Merging”, *preprint*, 2025.
- [3] **N. Verma**, K. Murray, K. Duh. “Merging feed-forward sublayers for compressed Transformers”, *preprint*, 2025.
- [4] **N. Verma**, M. Elbayad. “Merging text transformer models from different initializations”, *TMLR 2024 & High-dimensional Learning Dynamics (HiLD) @ ICML 2024*.
- [5] **N. Verma**, K. Murray, K. Duh. “Exploring geometric representational disparities between multilingual and bilingual translation models”, *LREC-COLING 2024*.
- [6] E. Salesky, **N. Verma**, P. Koehn, M. Post. “Multilingual Pixel Representations for Translation and Effective Cross-lingual Transfer”, *EMNLP*, 2023.
- [7] H.L. Xinyuan*, **N. Verma***, B. Odoom, U. Pradeep, M. Wiesner, S. Khudanpur. “JHU IWSLT 2023 Multilingual Speech Translation System Description”, *IWSLT*, 2023.
- [8] A. Hussein, C. Xiao, **N. Verma**, T. Thebaud, M. Wiesner, S. Khudanpur. “JHU IWSLT 2023 Dialect Speech Translation System Description”, *IWSLT*, 2023.
- [9] K. Marchisio, **N. Verma**, K. Duh, P. Koehn. “IsoVec: Controlling the Relative Isomorphism of Word Embedding Spaces”, *EMNLP*, 2022.
- [10] **N. Verma**, K. Murray, K. Duh. “Strategies for Adapting Multilingual Pre-training for Domain-Specific Machine Translation”, *AMTA*, 2022.
- [11] I. Li, J. Pan, J. Goldwasser, **N. Verma**, W. P. Wong, M. Nuzumlah, B. Rosand, Y. Li, M. Zhang, D. Chang, R. A. Taylor, H. M. Krumholz, D. Radev. “Neural Natural Language Processing for Unstructured Data in Electronic Health Records: a Review”, *Computer Science Review*, 2022.
- [12] L. Nan, C. Hsieh, Z. Mao, X. Lin, **N. Verma**, R. Zhang, W. Kryściński, N. Schoelkopf, R. Kong, X. Tang, M. Mutuma, B. Rosand, I. Trindade, R. Bandaru, J. Cunningham, C. Xiong, D. Radev. “FeTaQA: Free-form Table Question Answering”, *TACL*, 2021.
- [13] L. Nan, D. Radev, R. Zhang, A. Rau, A. Sivaprasad, C. Hsieh, X. Tang, A. Vyas, **N. Verma**, P. Krishna, Y. Liu. “DART: Open-domain structured data record to text generation”, *NAACL*, 2021.
- [14] R. Zhang, C. Westerfield, S. Shim, G. Bingham, A. Fabbri, W. Hu, **N. Verma**, and D. Radev. “Improving Low-Resource Cross-lingual document Retrieval by Reranking with Deep Bilingual Representations”, *ACL*, 2019.